



Benefits Basis of Estimate (BoE)
for
Aviation Safety Knowledge Management Environment
(ASKME)

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1.0 INTRODUCTION

This benefits Basis of Estimate (BoE) contains documentation of activities performed to assess the potential benefits of the Aviation Safety Knowledge Management Environment (ASKME). A draft version of this document was prepared for the Initial Joint Resources Council (JRC-2a) on May 22, 2003. This version is being prepared for the Final JRC-2b in early 2005.

1.1 Historical Background

ASKME is part of the AVR¹ National Enterprise Architecture and will serve Aircraft Certification Service (AIR) business applications. AIR assures safety standards related to the design and production of aircraft types. Other programs of the AVR National Enterprise Architecture include the System Approach for Safety Oversight (SASO), which is the Flight Standards Service (AFS) program, and the Aerospace Safety Information Management (ASIM) program—for the three remaining offices: Rulemaking (ARM), Aviation Medicine (AAM), and Accident Investigation (AAI).

All of these parts will be enabled by a common infrastructure called Regulation and Certification Infrastructure for System Safety (RCISS). The individual programs will also be linked so that they can share information.

The following are milestones of the ASKME program:

- The Mission Need Decision on 13 March 2002
- Quality Assurance Review #1 (QA-1) on 18 November 2002, which reviewed the benefits analysis methodology
- The benefits analysis for JRC-2a was completed on 21 February 2003
- The initial Requirements Document (iRD) was completed on 28 February 2003
- The initial Acquisition Program Baseline (APB) was finalized 24 March 2003
- Quality Assurance Review #2 (QA-2) on 22 April 2003, which reviewed the results and draft presentation of results for the JRC
- The JRC-2a on 22 May 2003
- Revised benefits analysis for JRC-2b in early December 2003
- Final Benefits Basis of Estimate (BoE) in December 2004

¹ Regulation and Certification line of business, Federal Aviation Administration.

1.2 Program Goals and Objectives

The goal of ASKME is to improve AIR's ability to fulfill its mission by enhancing the business process in AIR's various business activities. For more detail on AIR's mission, business activities, and strategic goals, and ASKME's goals and objectives to enhance that mission, see the Mission Need Statement (MNS).² However, there have been some changes since the publication of the MNS. All changes relevant to benefits are described in subsequent sections of this document.

While other potential areas were considered, the final estimate was based on safety benefits. The safety benefits are derived from the economic value associated with the prevention of accidents, including fatalities, injuries, and equipment loss and damage. In other words, if ASKME can facilitate a quicker and more comprehensive investigation of various data and identify potential weak links, some accidents may be avoided, resulting in a safety benefit.³ The estimates of the safety benefits are monetized, and these dollar values are used in the economic analysis (including the Net Present Value (NPV), Return on Investment (ROI), breakeven analysis, etc.).

The safety benefits are derived from the business process improvements associated with ASKME. These benefits are associated with an increase in the productive efficiency of the workforce due to the introduction of capital infrastructure. For example, with ASKME in place, investigators will be able to make a query about all the incidents related to a particular part under specific conditions more easily. Each query will take less time. The reduction in time per task due to information management and business process improvements is a productivity benefit.

Because improved productivity will allow a greater focus on safety issues, the productivity benefits will be realized indirectly through the safety benefits. The productivity benefits are quantified in terms of full time equivalents (FTEs), but they are not monetized via applying the labor rate because no labor force reduction is implied.

1.3 Structure and Terminology

In this document, it is important to distinguish between AIR functions and ASKME components. In designing ASKME, the work of AIR to meet its mission has been categorized into 11 interdependent **functional areas** and further subdivided into **sub-functions**. For more detail on sub-functions, see the iRD.⁴ The 11 functional areas are:

1. Rulemaking and Policy Development
2. Airworthiness Directive
3. Engineering Design Certification

² Mission Need Statement (MNS), Aviation Safety Knowledge Management Environment (ASKME), February 2002.

³ There were some benefits, anticipated in the Mission Need Decision, which will not be attributed to ASKME. For instance, increased collaboration with a decrease in travel requirements and Timely knowledge sharing with reduced travel. All benefits associated with travel reduction will be attributed to NEXGEN, not ASKME.

⁴ Some sub-functions have been reorganized since the printing of the iRD. All changes relevant to benefits are described in subsequent sections of this document.

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4. Production Approval and Certificate Management
5. Airworthiness Certification
6. Designee Management
7. Evaluation and Audit
8. External Inquiries
9. Investigation and Enforcement
10. Continued Operational Safety
11. International Coordination⁵

The AIR sub-functions will be *supported* by varying degrees by eight ASKME **components**. They are:

1. The Infrastructure (e.g., hardware servers)⁶
2. Electronic File System
3. Database Management System
4. Menu and Common Screen Format
5. Knowledge Transfer Environment (e.g., linkages between processes and databases)
6. Work-tracking Software
7. Electronic Forms
8. The Automated Workflow

A second distinction that should be explained is the difference between the fully supported and global sub-functions. With the exception of non-applicability⁷, there will be a select few AIR sub-functions that will be **fully supported** by ASKME via full development of all eight ASKME components.

However, most sub-functions will only be supported by the Infrastructure, Electronic File System, and Work Tracking Software components of ASKME. These are referred to as the **global** sub-functions.

An abbreviated version of the ASKME Requirements Matrix, showing the relationship between the ASKME components and AIR functional areas and sub-functions, is shown in Appendix A. The full version can be found in the IRD.

⁵ A detailed description of the functions of Air Certification can be found in Williams (1997).

⁶ The infrastructure component, common to ASKME, SASO, and the aviation medicine (“3rd box”) program, has been incorporated into RCISS.

⁷ For example, the Electronic Forms component is not relevant to the Continued Operational Safety functional area.

1.4 List of Participants

The following participants contributed to this benefits BoE.

Tony Dziepak, BP&D, Operations Research, benefits lead (from 21 September 2002)
Dave Chien, former ASD-430, benefits lead (through 20 September 2002)
Bryan Baszczewski, SETA-II, Operations Research, safety benefits
Chris Hunt, SETA-II, Operations Research, productivity benefits
Rakhee Sood, SETA-II, Operations Research, safety benefits
Steve Payne, Crown Consulting, supporting AVR (formerly AIR-520)

1.5 Time Frame

The deployment schedule entails the overlapping rollout of full support of two sub-functions per year beginning in FY 2004 and ending in FY 2009. The sub-function deployment is ordered based primarily on maximizing safety benefits. Thus, the sub-functions expected to yield the greatest safety benefits are deployed first. For benefits estimation purposes, the life cycle of ASKME ends in FY 2021.

2.0 SAFETY BENEFITS

2.1 Introduction

The ASKME Benefits Team extracted and analyzed fatal aviation accidents for commercial aviation, air taxi, general aviation, and foreign (FAR Parts 121, 135, 91, and 129 respectively) from the National Aviation Safety Data Analysis Center (NASDAC) database.⁸ In addition to the NASDAC, the Benefits Team researched literature from the “Commercial Airplane Certification Process Study”⁹ (the CPS “Blue Ribbon” Report) and the “Safer Skies Initiative”.¹⁰ Accidents were reviewed for their possible prevention through an ASKME program and, if chosen, assigned a probabilistic effectiveness rating that ASKME could have prevented the accident. The data was aggregated, quantified, and then projected over the life cycle of the ASKME program. Benefits were calculated using low, most likely, and high benefits estimates as inputs into a Monte Carlo simulation to develop high-confidence benefits. The benefits estimation methodology draws from concepts in the “Art of Aviation Safety Benefits Analysis.”¹¹

⁸ National Aviation Safety Data Analysis Center (NASDAC) databases available at <http://nasdac.faa.gov>.

⁹ Commercial Airplane Certification Process Study: An Evaluation of Selected Aircraft Certification, Operations, and Maintenance Processes,” available as a .pdf file on the International Federation of Airworthiness (IFA) website: <http://www.ifairworthy.org>.

¹⁰ “Safer Skies Initiative,” Federal Aviation Administration, April 1998; <http://www.faa.gov/index.cfm/apa/1267>.

¹¹ Stephen B. Cohen, “The Art of Aviation Safety Benefits Analysis,” October 2000, Rev. June 2001.

2.2 Data Collection

The team collected data from the NASDAC databases. The NASDAC database contains over 50,000 domestic and international aviation accidents and incidents from 1983-2002 of which approximately 9,000 contained one or more fatalities. To filter this dataset to more reasonable levels, the team focused on fatal accidents only. Next, the team consulted the National Transportation Safety Board (NTSB) Accident Cause Code Manual for possible causal factors for accidents, which might have been mitigated by ASKME capabilities.

The cause codes (see Appendix B) were selected to align with the functions of AIR. Some accident cause codes are broadly stated and may or may not provide accident data points attributable to AIR (e.g., “Information” may refer to “Information–Not Available from Manufacturer” or “Information–Not Issued by ATC Personnel”).

By limiting the data set to these cause codes, the team risked precluding ASKME-relevant accidents not containing these exact causal factors from the analysis. However, given the time and resource constraints, the size of the database, and the complexity of interpreting the data, the analysis demanded a strict data filtering process.

To mitigate the risk in this analysis, the benefits team also consulted the CPS “Blue Ribbon” Report for accident data points relevant to the ASKME program but not retrieved from the NASDAC database.

2.3 Data Review and Assignment

It is impossible for any one program to ensure 100% accident avoidance. Each accident has a history of interdependent causal factors contributing to the accident. Any one causal factor may be affected enough to prevent subsequent factors from causing an accident, therefore, each causal factor has a probability of accident avoidance. ASKME accident avoidance is defined as a probability that ASKME’s implementation may have prevented the accident through affecting one or more of the causal factors.

Assigning accident avoidance probabilities is a subjective process requiring informed opinions of subject matter experts with detailed knowledge of AIR processes and ASKME’s implementation of knowledge management, electronic file sharing, work tracking software, and training.

The first step in identifying ASKME-applicable historical accidents is determining which, if any, specific AIR sub-functions would contribute to the probable prevention of the accident given the new functionality of the ASKME program. In most cases of ASKME-applicable accidents, several sub-functions within AIR contribute to the possible prevention of an ASKME-applicable accident.

Once an accident is identified as ASKME-applicable, it was assigned a qualitative probability measure of Very High/High/Medium/Low/Very Low to denote the possible accident avoidance had ASKME been in place. Next, these ratings were assigned probabilities of .75, .60, .45, .30, and .15, respectively. Note that no probability is ever one--nothing occurs with certainty. Accidents not related to ASKME (probability=0) were separated from the pool of possible

accidents and excluded from further ASKME benefits analysis. The accident data points deemed applicable to ASKME and their respective ASKME effectiveness ratings are listed in Appendix C.

It is imperative to note that no matter how robust a safety program exists, poor judgment can be neither avoided nor anticipated. Our safety expert guidance found cases of poor judgment unavoidable with ASKME and the team excluded these data points from the analysis.

Since AIR operates at the domestic aircraft/component manufacturer level, the team excluded foreign-manufactured aircraft from the analysis and focused on domestic manufactured aircraft involved in both domestic and international accidents, meeting the data collection criteria.

2.4 Safer Skies Initiative

The ASKME program's improved data analysis, accident risk mitigation, and information sharing will complement the Safer Skies' initiative to reduce fatal commercial accidents by 80%. The ASKME program formally aligns to the Safer Skies agenda's policy that "Aviation safety is a shared responsibility." Within the Safer Skies agenda's shared safety policy, several aviation processes aligned with AIR functions, such as "safe airplane design, safety enhancing technology development, flight and maintenance operations, recommendations, documents, training, and support, maintenance planning, and safety related analysis", are identified as contributors to the reduction of fatal aviation accidents. The benefits team's accident data review shows that ASKME will contribute 3% of the identified preventable fatal Part 121 accidents in the Safer Skies' Commercial Aviation Safety Team's (CAST) agenda.¹²

2.5 Data Analysis and Quantification

2.5.1 ASKME Sub-function Accident Distribution

The team assigned benefits to each AIR sub-function assigned to ASKME-applicable accidents in the Data Validation phase as a sub-function necessary to prevent specific accidents. Each sub-function is weighted equally in their impact on the prevention of ASKME-applicable accidents. For a given accident, the sub-function attribution is divided equally among the sub-functions. For example, if four sub-functions are assigned to one accident, each sub-function receives 25% of the benefits for that accident. The accident allocation for each sub-function is the sum of the sub-function's accident attributions across all accidents.

¹² Commercial Aviation Safety Team (CAST) agenda; <http://www.faa.gov/index.cfm/apa/1054/FB4C3EF0-E880-4068-BE9B2E02A8F969E4>.

2.5.2 Historical ASKME-Related Statistics Calculations

The details of each applicable accident were entered into a spreadsheet and broken out by FAR Part and ASKME sub-function. The Office of Aviation Policy and Plans (APO) post-9/11 traffic growth projection was used to measure average annual historical accidents against operations and forecast future annual accidents. Table 1 describes the definitions and terminology used throughout this paper.

Table 1: Formulas For Historical Statistics

ASKME-Related Historical Statistics	Formula
Years Studied	(Maximum Accident Year – Minimum Accident Year) +1
Historical # of Applicable fatal Accidents by Sub-function	Count of all accidents in Years Studied by Sub-function Allocation
Sub-function Allocation by FAR Part	Sum(1/(Number of sub-functions identified as necessary to affect in the outcome of the accident)) across all accidents
Historical # of Fatalities	Sum of all Historical Fatalities in Years Studied
Historical # of Serious Injuries for Fatal Accidents	Sum of all Historical Serious Injuries in Years Studied
Historical # of Minor Injuries For Fatal Accidents	Sum of all Historical Minor Injuries in Years Studied
Average # Fatalities per accident w/o ASKME	Historical # of Fatalities/ Historical # of Applicable Accidents
Average # Serious Injuries per accident w/o ASKME	Historical # of Serious Injuries/ Historical # of Applicable Accidents
Average # Minor Injuries per accident w/o ASKME	Historical # of Minor Injuries/ Historical # of Applicable Accidents
Average Annual # Flights	(Total # of Flights in TAF between Minimum Accident Year and Maximum Accident Year) / (Years Studied)
Average Cost of Equipment Destroyed or Damaged per Accident	Sum of all Equipment Damage by Aircraft Type / Historical # of Applicable Accidents

2.5.3 Historical ASKME-Related Statistics

Table 2 summarizes ASKME-related historical statistics for each FAR Part as defined by the ASKME-Related Historical Statistics shown in Table 1.

Table 2: Historical Statistics By FAR Part

ASKME-Related Historical Statistics	Part 121	Part 135	Part 129	Part 91
Years Studied	1983-2002	1983-2000	1983-2002	1983-2000
Historical # of Applicable Fatal Accidents	10	8	1	6
Historical # of Fatalities	704	58	2	9
Historical # of Serious Injuries for Fatal Accidents	68	16	60	3
Historical # of Minor Injuries for Fatal Accidents	162	9	96	0
Average # Fatalities w/o ASKME, per accident	70.4	7.2	2.0	1.5
Average# Serious Injuries w/o ASKME for Fatal Accidents, per accident	6.8	2.0	60	0.5
Average# Minor Injuries w/o ASKME for Fatal Accident, per accident	16.2	1.1	96	0
Average Cost of Equipment Destroyed or Damaged per accident	\$ 9.8M	\$ 0.47M	-	\$ 0.55M
Average Annual # of Fatal Accidents w/o ASKME	0.50	0.44	0.05	0.33

2.5.4 Predicted ASKME-Related Statistics

The following predicted ASKME-related statistics were calculated for each sub-function by FAR Part per year. Table 3 shows each statistic and the formula used to compute that statistic.

Table 3: Formulas for Predicted Statistics

ASKME-Related Predicted Statistics	Formula
Average Accident Rate Reduction	$\frac{p(s)_1 + p(s)_2 + \dots + p(s)_n}{n}$, where p(s) _x is the probability of accident avoidance given ASKME for accident 'x.', based on expert opinion, and n=number of accidents
Predicted Number of Accidents w/o ASKME	Avg. Annual # Historical Accidents • APO post-9/11 Growth Rate for Year X
Predicted Number of Accidents with ASKME	(1-Avg. Accident Rate Reduction) • Predicted # of Accidents w/o ASKME
Predicted # of Accidents Avoided with ASKME	Avg. Accident Rate Reduction • Predicted # of Accidents w/o ASKME • ASKME Development Schedule
Predicted # of Fatalities w/o ASKME	Avg. per Accident # of Fatalities w/o ASKME • Predicted # of Accidents w/o ASKME
Predicted # of Injuries w/o ASKME	Avg. per Accident # of Injuries w/o ASKME • Predicted # of Accidents w/o ASKME
Predicted # of Fatalities Avoided with ASKME	Avg. per Accident # of Fatalities • Prevented # Accidents with ASKME
Predicted # of Injuries Avoided with ASKME	Avg. per Accident # of Injuries • Prevented # Accidents with ASKME
Predicted Cost of Equipment Destroyed or Damaged	Predicted # of Accidents Avoided with ASKME • Average Cost of Equipment Destroyed or Damaged

The Average Accident Rate Reduction, as described in Table 3, by FAR Part is listed in Table 4.

Table 4: ASKME Average Accident Prevention Effectiveness Ratings

FAR Part	ASKME-Related Average Accident Rate Reduction
121	0.24
135	0.23
91	0.30
129	0.45

The team used APO's 2003 traffic growth rates to predict the number of accidents without ASKME between 2004 and 2020. The projection in 2021 used a Microsoft Excel straight line forecast based solely on the APO 2004-2020 projections.

2.5.5 High, Most Likely, and Low Benefits Estimates Methodology

For all FAR Parts by sub-function, applying the highest accident avoidance probability (.75) to the entire pool of ASKME-applicable accidents provides a high benefits estimate on the total pool of possible ASKME benefits. Conversely, applying the least likely probability of accident avoidance (0) to the entire pool of ASKME-applicable accidents provides a lower bound of 0 on the pool of possible benefits. The most likely benefit estimates applies the average accident rate reduction value as described above in Table 4. These three benefits point estimates form a triangular distribution for the high-confidence benefits analysis.

2.6 Benefits Calculation

2.6.1 Economic Values

Standard economic values for fatalities, serious injuries, and minor injuries were applied to the Predicted Number of Fatalities/Injuries avoided for each year for the ASKME program:

- Fatality: \$3,000,000
- Serious Injury: \$172,500
- Minor Injury: \$6,000

Standard economic values for equipment replacement and damage were applied to each accident according to the type of aircraft involved in the accident. Equipment replacement economic values assign the value of identical or nearly identical aircraft. Restoration economic values assign the cost of repairable damage to substantially damaged equipment. Minor damage is generally a marginal cost and is not reported in the economic values.

2.6.2 Benefits Dependency on ASKME Deployment Schedule

The benefits stream is dependent on the phase-in schedule of the sub-functions and components. ASKME is to be deployed in the following manner:

- The global components will be phased in over several years; they will be fully operational in FY 2011.
- The sub-functions will be phased in independently of each other but dependent on the Historical Scanning Activities and all fully implemented in FY 2011.

Table 5 illustrates the phase-in of the global components and sub-functions. The shaded boxes represent the years in which the components will be developed.

In addition to the deployment schedule, the ASKME Team used assumptions about a benefits lag to account for “learning curve” and training effects. In general, for the fully supported sub-functions, the team assumed 33.3% potential on the full year after deployment, 66.7% two years after, and 100% beginning in the third year after deployment. This basic assumption about the phase-in schedule is the same as that used with SASO.¹³ This is also consistent with similar federal government information technology and business process improvement programs, such as CODIS.¹⁴ Partial year implementations are prorated accordingly.

The safety benefits for ASKME’s predicted accident prevention are dependent on the ASKME deployment schedule. Given that ASKME is comprised of a set of AIR sub-functions, the benefits are measured at the individual AIR sub-function level. Each sub-function’s benefit stream depends upon two factors: 1) the development schedule and, 2) the degree to which the ASKME global component *Electronic File System’s* task of *Historical Scanning Activities* is completed. Each sub-function identified must have *Historical Scanning Activities* fully implemented to achieve 100% of their benefits. Sub-function 2A, *Airworthiness Directive Development*, phases in more gradually (i.e., greater than a three-year baseline) than others due to its deployment schedule’s concurrent relationship to the deployment schedule of the *Electronic File System’s* task of *Historical Scanning Activities*.

This means that the benefits for all sub-functions that are deployed subsequent to the *Historical Scanning Activities* deployment are ramped up at 33%, 67%, and 100% for the first, second, and third (and all subsequent) years after deploy, adjusted for partial-year implementations. However, sub-functions that are deployed prior to the year of *Historical Scanning Activities* will be ramped up more slowly (i.e., Sub-function 2A – AD development). After working meetings with representatives from AIR, the final ramp-up phase-in multipliers are given in Table 5.

¹³ See Section 2.7.2.

¹⁴ Combined DNA Index System (CODIS), Federal Bureau of Investigations (FBI).

Table 5: Sub-function Phase-In And Effects On Benefits

Code	Sub-function	FY 04	FY 05	FY 06	FY 07	FY 08	FY 09	FY 10	FY 11	FY 12	FY 13-21
	Global components (includes Historical Scanning Activity)			.026	.138	.364	.676	.893	1.0	1.0	1.0
2a	AD development			.104	.375	.708	1.0	1.0	1.0	1.0	1.0
10a	Monitor safety related data					.167	.500	.833	1.0	1.0	1.0
3a	Design approval (TC/STC)				.333	.667	1.0	1.0	1.0	1.0	1.0
10b	Assimilate lessons learned from accidents						.333	.667	1.0	1.0	1.0
9a	Compliance and enforcement actions					.333	.667	1.0	1.0	1.0	1.0
5a	Airworthiness Certification						.333	.667	1.0	1.0	1.0
6a	Supervision/ past performance						.333	.667	1.0	1.0	1.0

2.6.3 Global Components

The safety benefits analysis also captured quantifiable benefits from the global components *Electronic File System* and *Work Tracking Software* applications in the sub-functions Rulemaking and Policy Development and Airworthiness Directive–Compliance. These two sub-functions are not fully supported; however, the two ASKME global components help achieve these benefits through the accessibility of documentation and workflow available from these sub-functions. The global components’ benefits accrual depends on two factors: the number of sub-functions fully implemented with ASKME and the number of documents scanned.

2.6.4 Benefits Accrual--Delayed Sub-functions

The following two sub-functions listed in Table 6 require a four-year phase-in of the benefits due to their concurrent deployment with the *Historical Scanning Activities* deployment and/or due to a partial first year implementation.

Table 6: Causes of Delayed Deployment for Affected Subfunctions

Code	Sub-function	Dependency on Historical Scanning Activities	Partial Year Implementation
2a	Airworthiness Directive – Development	Yes	Yes
10a	Continued Operational Safety – Monitor Safety Related Data	No	Yes

In addition to a partial first year implementation, the first two years of deployment for *Sub-function 2a – Airworthiness Directive Development* receives a penalty equal to the level of estimated activity to be completed for the *Historical Scanning Activities*. The *Sub-function 10a - Continued Operational Safety – Monitor Safety Related Data* deploys in mid-year FY 2008 and receives only half of the standard benefits accrual rate 33% for its initial year.

2.6.5 Benefits Accrual--Baseline Sub-functions

The following five sub-functions accrue benefits at the baseline rate of 33% per year in the first year following the last year of the sub-function's development until 100% of the sub-function's benefits are realized in the third year:

- Design Approval (TC/STC)
- Assimilate Lessons Learned from Accidents
- Compliance and Enforcement Actions
- Standard Airworthiness Certification
- Supervision/Past Performance

2.6.6 Benefits Period

Using the above deployment constraints on each ASKME sub-function, benefits were calculated for each applicable year for each Low Estimate, Most Likely, and High Estimate scenario. The applicable years for benefits calculations follow the ASKME Life Cycle schedule from 2004-2021.

2.6.7 Predicted Avoided Fatalities and Equipment Replacement/Damage Benefits Estimates

The predicted number of avoided fatalities by FAR Part for each benefits point estimate is listed in Table 7. The benefits point estimates use accident effectiveness ratings of 0%, Average of Team Inputs, and 75% for Low, Most Likely, and High Estimates, respectively (see Table 4 (above) for more detail on the Average of Team Inputs). Tables 8 and 9 highlight the number of avoided fatalities and avoided accidents, respectively, by benefits point estimate between 2004 and 2021.

Table 7: Avoided Fatalities, 2004-2021

Part	Low Estimate	Most Likely	High Estimate
121	0	138	432
135	0	11	38
91	0	4	9
129	0	0.7	1
Totals:	0	154	481

Table 8: Avoided Injuries, 2004-2021

Part	Low Estimate	Most Likely	High Estimate
121	0	45.2	141.2
135	0	5.0	16.5
91	0	1.2	3.1
129	0	59.7	99.6
Totals:	0	111	260

Table 9: Avoided Accidents, 2004-2021

Part	Low Estimate	Most Likely	High Estimate
121	0	2.0	6.1
135	0	1.6	5.3
91	0	2.5	5
129	0	0.4	0.6
Totals:	0	6	18

The estimated benefits (Constant \$M) resulting from predicted avoided fatalities and injuries between 2004 and 2021 are listed by FAR Part and benefits point estimate in Table 10.

Table 10: Fatalities/Injuries Avoided Benefits, 2004-2021 (Constant \$M)

Part	Low Estimate	Most Likely	High Estimate
121	0	\$418	\$1304
135	0	\$35	\$117
91	0	\$11	\$28
129	0	\$6	\$11
Totals:	0	\$470	\$1460

The estimated benefit (constant \$M) resulting from predicted avoided equipment replacement and damage between 2004 and 2021 is listed by FAR Part and benefits point estimate in Table 11.

Table 11: Avoided Cost of Equipment Replacement and Damage Benefits, 2004-2021 (Constant \$M)

Part	Low Estimate	Most Likely	High Estimate
121	0	\$19.3	\$60.1
135	0	\$0.8	\$2.5
91	0	\$1.4	\$3.4
129	0	N/a	N/a
Totals:	0	\$21.4	\$66

2.6.8 High-Confidence Benefits Results

The high-confidence benefits were generated using the common Monte Carlo simulation package “Crystal Ball” and Microsoft Excel. The benefits team input the monetized low, most likely, and high benefits point estimates for each sub-function into the simulation as triangular distributions to generate a 20th percentile high-confidence constant dollar value.

A correlation factor of 0.6 between each sub-function is used in the Monte Carlo simulation. This is based on the accident review process, which suggested there was some (but not complete) dependency between sub-functions in achieving the projected safety benefits. The resulting distribution of the simulation, in 5-percentile increments, is listed in Table 12.

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Table 12: Tabular Depiction of Monte Carlo Simulation Results

Monte Carlo Distribution	
<i>Percentile</i>	<i>Value</i>
0	\$ 24,597,438
5	\$ 271,936,181
10	\$ 343,219,388
15	\$ 396,892,555
20	\$ 440,787,165
25	\$ 480,338,197
30	\$ 517,140,468
35	\$ 552,400,982
40	\$ 586,743,959
45	\$ 622,047,646
50	\$ 656,428,554
55	\$ 692,411,397
60	\$ 729,851,186
65	\$ 770,176,566
70	\$ 811,509,905
75	\$ 857,153,936
80	\$ 908,534,120
85	\$ 967,076,681
90	\$ 1,038,690,829
95	\$ 1,139,990,405
100	\$ 1,484,281,801

The distribution is also plotted as a cumulative distribution in Figure 1.

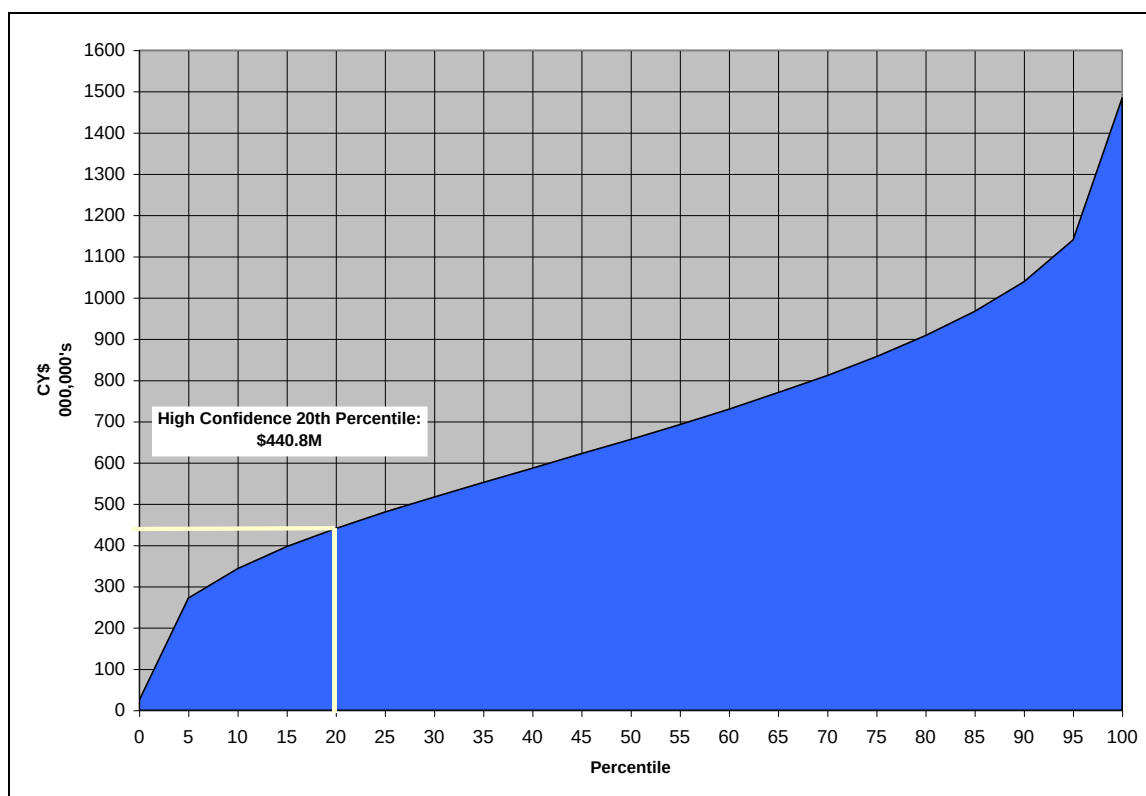


Figure 1: Cumulative frequency of Monte Carlo Simulation Results

Using the 20th percentile value of the simulation result, the benefits team back-allocated the ratio of the 20th percentile to the Most Likely value to calculate the high-confidence metrics necessary to achieve the high-confidence benefits. The high-confidence metrics and benefits for each FAR Part are listed in Tables 13 and 14, respectively.

Table 13: High-Confidence Metrics, 2004-2021

FAR Part	Accident Prevention Effectiveness Rating	Predicted ASKME-Related Fatal Accidents (2004-2021)	Predicted ASKME-Related Fatalities (2004-2021)	Predicted Avoided Accidents	Predicted Avoided Fatalities	Predicted Avoided Injuries (Serious & Minor)
121	21%	11.2	706	1.8	124.0	40.5
135	20%	11.5	75	1.4	10.3	4.4
91	27%	13.4	18	2.2	3.3	1.1
129	40%	1.1	2	0.3	0.7	53.6
Totals:	-	37.3	801	5.7	138.3	99.6

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Table 14: High-Confidence Benefits, 2004-2021, (\$M)

FAR Part	Avoided Fatalities Benefits	Avoided Injuries (Serious & Minor) Benefits	Avoided Equipment Damage Benefits	Total Benefits
121	\$ 372.1	\$ 2.2	\$ 17.3	\$ 391.6
135	\$ 30.9	\$ 0.5	\$ 0.7	\$ 32.1
91	\$ 9.9	\$ 0.2	\$ 1.2	\$ 11.3
129	\$ 2.1	\$ 3.7	-	\$ 5.8
Totals:	\$ 414.9	\$ 6.7	\$ 19.1	\$ 440.8

The benefits team also calculated the high-confidence benefits achieved by FAR Part for each sub-function. The high-confidence benefits (constant \$M) by ASKME sub-function are given in Table 15.

Table 15: ASKME High-Confidence Safety Benefits By Subfunction

Function	Sub-function	Part 121	Part 135	Part 91	Part 129	Totals by Sub-function
#10 Continued Operational Safety	Assimilate Lessons Learned from Accidents	\$103.0	\$5.6	\$1.8	\$1.8	\$112.2
#2 Airworthiness Directive	Development	\$97.2	\$9.1	\$3.0	\$0.0	\$109.4
#3 Engineering Design Certification	TC/STC Eng. Des. Appr., Amd., and Cert. Mgmt.	\$74.4	\$4.3	\$2.4	\$2.1	\$83.3
#10 Continued Operational Safety	Monitor Safety Related Data	\$67.5	\$5.8	\$2.2	\$1.9	\$77.3
#5 Airworthiness Certification	Standard Airworthiness Certification	\$29.1	\$2.8	\$1.8	\$0.0	\$33.7
#1 Rulemaking and Policy Development	Rulemaking Projects	\$20.5	\$0.0	\$0.0	\$0.0	\$20.5
#2 Airworthiness Directive	Compliance	\$0.0	\$2.5	\$0.0	\$0.0	\$2.5
#9 Investigation and Enforcement	Compliance and Enforcement Actions	\$0.0	\$2.0	\$0.0	\$0.0	\$2.0
Totals by FAR Part:		\$391.6	\$32.1	\$11.3	\$5.8	\$440.8

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The annualized high-confidence benefits, as provided to the APB, are listed in Table 16.

Table 16: Annualized High-Confidence Benefits

Annual Constant High-Confidence Benefits	
<i>Year</i>	<i>Value</i>
2006	\$ 707,684
2007	\$ 4,361,830
2008	\$ 9,704,678
2009	\$ 19,532,492
2010	\$ 25,883,798
2011	\$ 31,200,405
2012	\$ 31,857,522
2013	\$ 32,524,499
2014	\$ 33,202,296
2015	\$ 33,886,878
2016	\$ 34,581,957
2017	\$ 35,282,884
2018	\$ 35,988,319
2019	\$ 36,699,682
2020	\$ 37,420,163
2021	\$ 37,952,078
Total	\$ 440,787,165

The magnitude and effect of each sub-function on the Most Likely value is shown below in a Tornado Chart in Figure 2. This analysis determines which sub-functions are most likely to have the greatest impact on the benefits independent of the influence of other sub-functions. The Most Likely value is tested for its sensitivity to each individual sub-function without any correlation factors. Figure 2 shows the Most Likely benefits value being most influenced by the top sub-functions and least influenced by the bottom sub-functions.

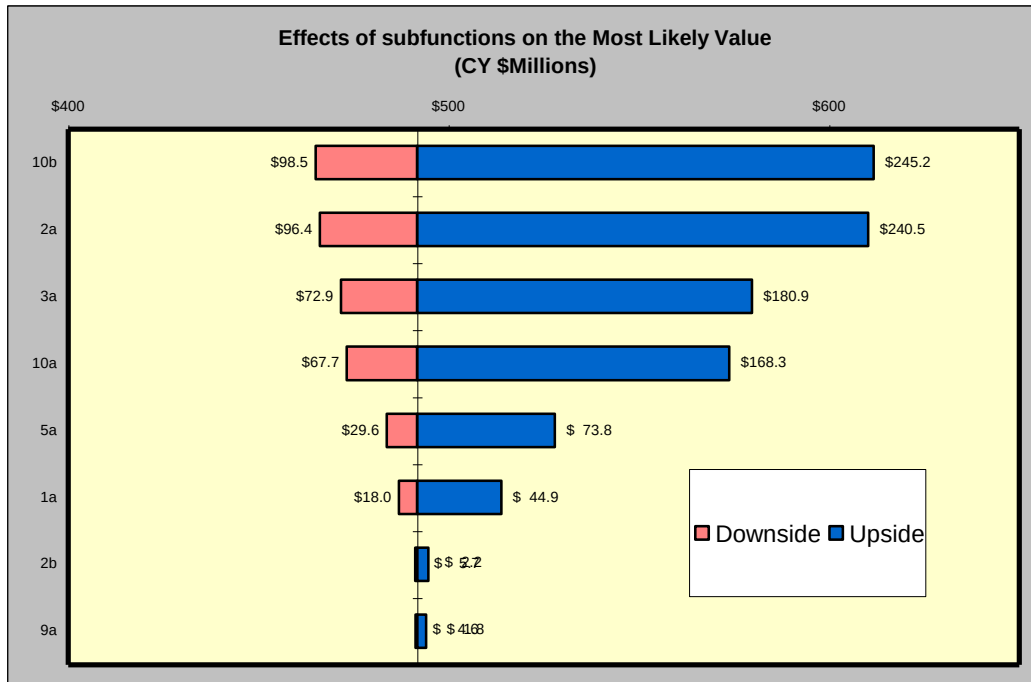


Figure 2: Effects of Sub-functions on the Most Likely Value

3.0 PRODUCTIVITY BENEFITS

3.1 Introduction

Productivity benefits include quantitative and qualitative benefits associated with improvements from business process re-engineering and information management. The business process will be improved through automation and electronic storing and sharing of information. Data will become standardized, and accessible through several networked servers. In addition, an automated workflow system, similar to the Project Activity Folder (PAF),¹⁵ will be applied to some of the ASKME sub-functions. More detailed descriptions are given in the iRD.

3.2 Methodology¹⁶

The productivity benefits are quantified in terms of FTE years—a flow variable approximating the work capacity of a technical employee for a year.¹⁷⁸ This is calculated by simply dividing the total annual direct technical hours by 1,676, which is AIR's value for the average number of hours *per annum*.

The benefits team has decided not to put a monetary value on the FTE years because AIR will not realize this benefit directly through staff reduction. Rather, the labor hours freed from the

¹⁵ Project Activity Folder (PAF) Function Automation Project Concept Paper, Oct 2000.

¹⁶ The foundation of this structure was an adaptation to ASKME of material in the benefits section of the Investment Analysis Workshop.

¹⁷ This is in contrast to FTEs, which is a stock variable measuring the number of employees on hand at a given point in time.

AIR staff by the ASKME productivity gain will be applied to additional work. This will not be a replacement for some of the work currently completed by designees--this is new work that is not currently being done. With ASKME, ASIs and ASEs will spend less of their time on administrative tasks and more of their time on improving safety.

The quantification of productivity benefits in FTE years allows us to measure the relative magnitude of the shift from administrative tasks to a more safety focus that ASKME will provide. The nature of this new work includes completing backlogged assignments, doing activities in a more complete, comprehensive and proactive manner, and/or additional research.

The productivity benefits will be realized indirectly through the safety benefits monetized in Section 2. Because the nature of this work differs from both the current assignments of AIR and the designee activities, it would be misleading to monetize the FTE years using either the FAA employee technical wage rate or the designee wage rate.

The productivity benefits were quantified by analogy to productivity gains made by similar programs in the past. The team relied heavily on two studies: the Regulatory Guidance Library/Designee Information Management (RGL/DIN) Evaluation¹⁸ and PAF. The RGL/DIN evaluation was a retrospective cost-benefit analysis prepared in 2000. The RGL is an Internet-publicly-accessible database of current and historical aviation regulatory and guidance documents, implemented in 1998. The DIN system was also implemented in 1998. It is an internal system that replaced several earlier databases. It assists designees working on behalf of both AIR and Flight Standards (AFS).

The RGL/DIN Evaluation determined that there were productivity gains of as much as 87% in the completion of certain tasks as a result of these databases. Because ASKME already has a higher baseline, the team does not expect the productivity gain to be as dramatic.¹⁹ However, we can still use the analogy to calibrate an estimation of the benefits. In other words, while the magnitude may not be the same, the good conceptual fit enables us to estimate productivity gain with less risk.

3.2.1 Determination of benefit categories

The ASKME team produced a benefits summary table indicating which ASKME functions and sub-functions are expected to have productivity benefits. This document often refers to sub-function by a number-letter combination. Table 17 translates the fully supported sub-functions (Column 2) and functional areas (Column 3) by alphanumeric code (Column 1).

¹⁸ Manganaro, Paul, "Aircraft Certification Service Evaluation of the Regulatory Guidance Library (RGL) and Designee Information Management (DIN) Information Systems," February 2001.

¹⁹ That is, AIR's current mode of operations without ASKME is more advanced than it was than the baseline was before RGL/DIN was adopted.

Table 17: Expected Productivity Gain Factors By Sub-function

Code	Sub-function	Functional area	FY 2002 ASE hours	FY 2002 ASI hours
2a	AD development	Airworthiness directive	47,922	0
10b	Assimilate lessons learned from accidents	Continued operational safety	6,514	150
3a	Design Approval (TC/STC)	Engineering design certification	392,327	9169
10a	Monitor safety related data	Continued operational safety	6,514	150 ²⁰
5a	Airworthiness certification	Airworthiness certification	0	14,211
9a	Compliance and enforcement actions	Investigation and enforcement	8,265	11,979
6a	Supervision/past performance	Designee management	22,080	30,906

3.2.2 Establish the baseline

The ASKME Team determined the current mode of operations through several interviews with field specialists. The following is sample of the questions asked:

- What methods are used to store the information?
- Is the information entered into a database, paper form, or does it reside only in the minds of engineers and inspectors?
- What methods are involved in retrieving the information? Specifically, can a person simply search a database, or must it be retrieved by microfiche, by paper file, or by personal interview? Do not spare any detail.

From these interview questions, the team was able to assess which analogy was the best fit to the specific sub-function. Then by comparing the difference between the baseline and the ASKME enhancement, the team assigned a productivity gain factor (a percentage) to the sub-function. During the JRC-2b phase, the ASKME team conducted several more interviews with subject matter experts through emails, phone calls, teleconferences, and site visits to further refine our productivity factors.

Workload estimation: The benefits team received AIR annual reports for fiscal years 1996-98, and 2000-02. From this detailed data, the team partitioned the total direct technical hours (including pilots, engineers, and inspectors) across sub-functions according to the FY 2002 workload.²¹ Table 18, Columns 4 and 5, show the total FY 2002 Engineers' and Inspectors' hours assigned to each sub-function.

²⁰ Hours for sub-functions 10a and 10b could not be separated. The total hours for 10a and 10b are 16,155. The number is divided equally across the two subfunctions. This does not affect our estimate because we estimate the same low- best- and high estimates for productivity gain.

²¹ This data does not include designee hours.

Annual growth estimation: The benefits team took the total direct technical hours from the available AIR annual reports, which included FY 1996-98, and 2000-01, plus the 1998 estimate for 1999. The best-fit exponential growth curve to these six plot points indicates an estimated annual workload increase of about 2.1%. The team used this curve as a basis for forecasting the baseline total direct technical hours for the ASKME program life cycle.

The historical and forecast total direct technical hours are illustrated in Figure 3. The five actual total direct technical work hours are shown as solid (blue) points, and the FY 1998 estimate for FY 1999 is shown as a hollow (red) point. The best-fit exponential growth curve of 2.09% is the solid (yellow) line with individual points. A high and low range is represented by dashed (magenta and green) lines, respectively.

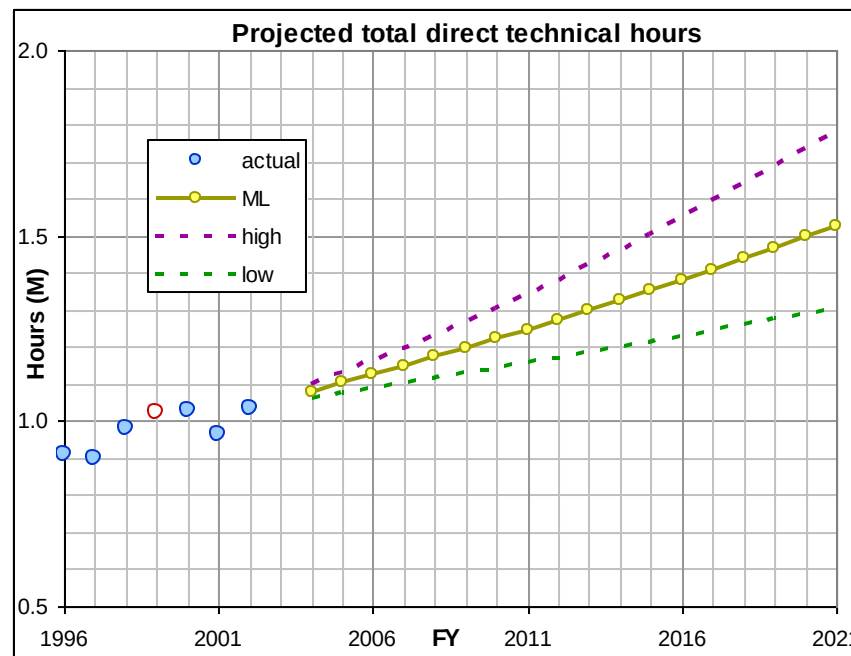


Figure 3: A Comparison of Growth Rates

3.2.3 Anticipated benefits

Through several working meetings, teleconferences, phone interviews, and field trips with experts in the field, the ASKME team agreed on high-end, low-end, and most-likely estimates for productivity gain factors. In most cases, work category hours assigned to each sub-function had several different distributions.

Table 18, columns 3-5, shows the final productivity percentage estimates by sub-function for all of the fully supported sub-functions. In addition, the team expects to receive some productivity gain in the globally supported sub-functions.

Table 18: Distribution of ASE/ASI Hours Across Sub-functions

Code	Sub-function	Low	Most Likely	High	FY 2002 ASE hours	FY 2002 ASI hours
2a	AD development (all)	17%	24%	32%	47,922	0
10ab	Continued operational safety	12%	16%	20%	8382 ²²	0
10ab	Continued operational safety	2%	3.5%	5%	4646	0
10ab	Continued operational safety	4%	8%	10%	0	299
10ab	TOTAL				13,028	299
3a	Design Approval	17%	24%	32%	328,892	0
3a	Design Approval	2%	3.5%	5%	15,376	0
3a	Design Approval	0%	0%	0%	48,059	0
3a	Design Approval	15%	19%	22%	0	175
3a	Design Approval	17%	25%	32%	0	8994
3a	TOTAL				392,327	9,169
5a	Airworthiness certification	17%	25%	32%	0	188
5a	Airworthiness certification	0%	0%	0%	0	5,116
5a	Airworthiness certification	15%	19%	22%	0	3,628
5a	Airworthiness certification	20%	28%	35%	0	2,800
5a	Airworthiness certification	22%	26%	30%	0	2,153
5a	Airworthiness certification	14%	17%	20%	0	326
5a	TOTAL				0	14,211
9a	Compliance/enforcement	12%	16%	20%	8,265	2,031
9a	Compliance/enforcement	27%	32%	35%	0	1,632
9a	Compliance/enforcement	20%	24%	27%	0	8,316
9a	TOTAL				8,265	11,979
6a	Supervision/past performance	5.8%	8.5%	11.5%	22,080	0
6a	Supervision/past performance	12%	16%	19%	0	21,151
6a	Supervision/past performance	12%	15%	18%	0	6,709
6a	Supervision/past performance	10%	12%	13%	0	2,448
6a	Supervision/past performance	4%	6%	7%	0	598
6a	TOTAL				22,080	30,906
	global	2%	3.5%	5%	258,633	140,644
	global	25%	25%	24%	4,225	0
	global	0%	0%	0%	16,520	542
	global	5%	6%	7%	0	24,810
	global	3%	5%	6%	0	4,599
	global	5%	6.5%	8%	0	24,286
	global	10%	12.5%	15%	0	5,700
	global	1%	1.5%	2%	0	3,639
	global TOTAL				279,378	204,220
	Other (all)	0%	0%	0%	1,437	0
	GRAND TOTAL				764,437	270,784

²² Hours for subfunctions 10a and 10b could not be separated. The total hours for 10a and 10b are 16,155. The number is divided equally across the two subfunctions. This does not affect our estimate because we estimate the same low- best- and high estimates for productivity gain.

The Team used the AIR FY 2002 annual report to determine the distribution of direct technical hours across AIR work categories. Most of these work categories matched with the ASKME sub-functions. Some AIR work categories did not match with a particular ASKME sub-function where there was more than one AIR work category. The final distribution of FY 2001 ASE and ASI direct technical hours across sub-functions are shown in Table 18, columns 6 and 7.

The hours that fell into the global sub-functions were assigned an expected productivity factor of 1.5%, with 1.0% for the low end and 2.5% for the high end. In FY 2001, there were 474,134 such hours. Other hours, a total of 3,176, were assigned no productivity gain.

The ASKME post-acquisition mode of operation was determined by the components' contribution of support to the sub-function. A percentage in time reduction was assigned to each sub-function based on the extent of the enhancement.

The team assigned a productivity factor through analogy with similar projects, such as RGL and PAF. A saving of 87%, similar to RGL, would derive if a sub-function was taken from a combination of a paper, microfiche, and commercial sources to a fully searchable and (secure) accessible database. Most sub-functions would have fewer timesavings because the baseline is more advanced, and/or the sub-function is only globally supported.

Two assumptions that were applied to the safety benefits also apply to productivity benefits: the same phase-in multipliers and the same 60% correlation factor between sub-functions. A Monte Carlo simulation was run with Crystal Ball to generate the high-confidence (20th percentile) figures.

3.3 Results

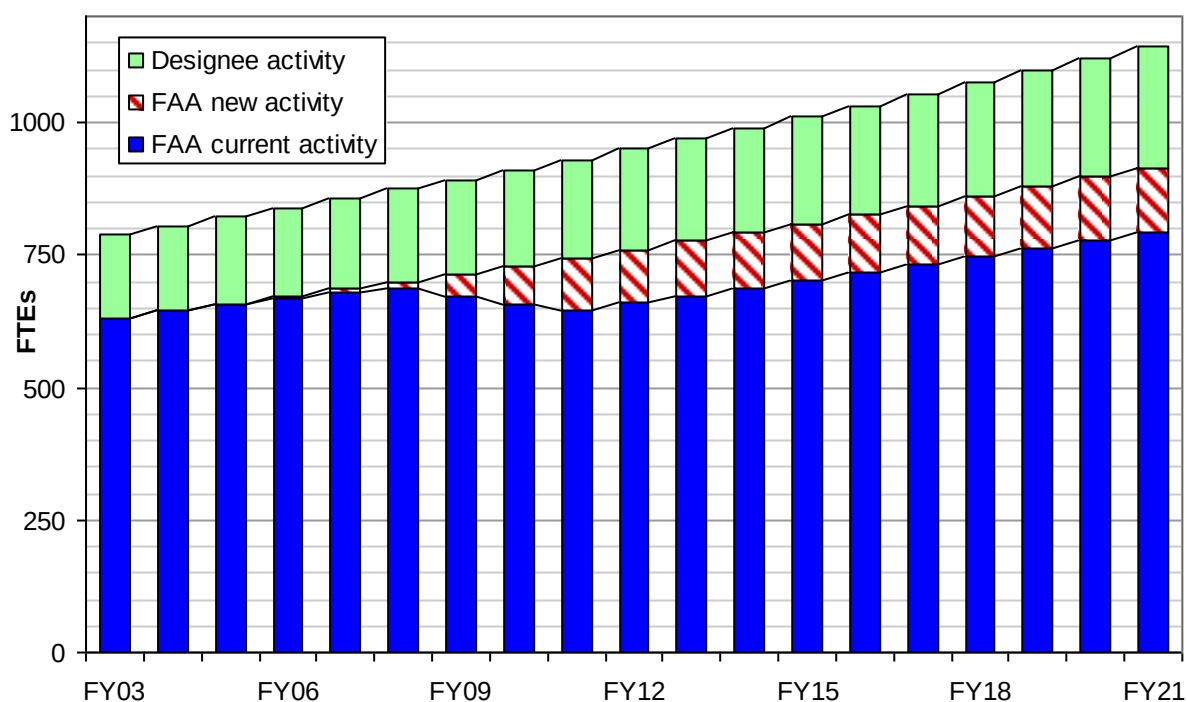
Over the life cycle of ASKME, AIR is expected to gain approximately 1,140 FTEs of productivity. The gains are illustrated by year in Table 19. Column 2 shows the baseline workload growth without ASKME, based on the 2.09% expected annual growth in workload. Column 3 shows the amount of FTEs required to do the same amount of work with ASKME, and Column 4 is the productivity gain—the difference between Columns 2 and 3. The Total row (bottom, in bold) represents the cumulative FTEs, which is the sum of all FTEs over all years.

Table 19: FTE-Years and FTEs by Year

FY	Baseline FTEs	With ASKME	Productivity gain
2005	657.2	657.2	0.0
2006	671.0	670.0	1.0
2007	685.0	680.8	4.2
2008	699.3	689.0	10.3
2009	713.9	676.6	37.3
2010	728.8	667.3	61.5
2011	744.1	659.9	84.2
2012	759.6	673.7	85.9
2013	775.5	687.8	87.7
2014	791.7	702.1	89.6
2015	808.2	716.8	91.4
2016	825.1	731.8	93.3
2017	842.4	747.1	95.3
2018	860.0	762.7	97.3
2019	878.0	778.7	99.3
2020	896.3	794.9	101.4
2021	915.0	811.5	103.5
Total	13251	12108	1143

The productivity gain is also represented in graphical form in Figure 4. The FAA activity is currently shown in blue/black. The team is uncertain of the exact amount of designee activity, however, the amount over and above the FAA activity, as shown in green/light gray.

ASKME productivity gains are represented with (red and white) crosshatched areas between the FAA workload and the designee workload.

**Figure 4: Graphical Representation of ASKME Productivity Gains by Year**

Finally, Table 20 shows a proportional breakdown of the high-confidence productivity gain by sub-function.

Table 20: Productivity Gain by Sub-function (in FTEs)

Code	Sub-function description	FTEs
2a	AD development	98.1
10a&10b	Continued operational safety	13.6
3a	Design approval	733.3
5a	Airworthiness certification	18.4
6a	Supervision/past performance	56.7
9a	Compliance/enforcement	34.8
Global		188.3
Total		1143

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APPENDICES

Appendix A: Abbreviated ASKME Requirements Matrix

Table A-1: Abbreviated ASKME Requirements Matrix

Legend		ASKME Components	8: Automated Workflow	7: Electronic Forms	6: Work Tracking Software	5: Knowledge Transfer Environment	4: Menu and Common Screen Format	3: DBMS	2: Electronic File System	1: Infrastructure												
Unsupported																						
Irrelevant																						
Global																						
ASKME																						
Subcomponents		Applications		Routing	Approval	Monitor (Status)	Manual Means	Form Prep	Recording resource alloc/ utiliz & generating reports	Link between processes	Links to existing AVR apps & databases	Classes and Seminars	Lessons Learned & Best Practices	CHI	Output Standard	Database Mgt System	Document scanning	Store/retrieve historical info	Primary Servers	Secondary Servers	Network Connections at 37 Sites	ASKME-Specific HW, SW
Sub-functions		Applications		Routing	Approval	Monitor (Status)	Manual Means	Form Prep	Recording resource alloc/ utiliz & generating reports	Link between processes	Links to existing AVR apps & databases	Classes and Seminars	Lessons Learned & Best Practices	CHI	Output Standard	Database Mgt System	Document scanning	Store/retrieve historical info	Primary Servers	Secondary Servers	Network Connections at 37 Sites	ASKME-Specific HW, SW
AIR Functions		Applications		Routing	Approval	Monitor (Status)	Manual Means	Form Prep	Recording resource alloc/ utiliz & generating reports	Link between processes	Links to existing AVR apps & databases	Classes and Seminars	Lessons Learned & Best Practices	CHI	Output Standard	Database Mgt System	Document scanning	Store/retrieve historical info	Primary Servers	Secondary Servers	Network Connections at 37 Sites	ASKME-Specific HW, SW
1: Rulemaking and Policy Development Activity	a. Rulemaking Projects (CFR)																					
	b. Orders																					
	c. Notices																					
	d. Policy Statements																					
	e. Advisory Circulars																					
	f. Technical Std Orders																					
	g. Exemptions																					
	h. Special Conditions																					
2: Airworthiness Directive	a. Development																					
	b. Compliance																					
3: Engineering Design Certification	a. TC/STC Eng Design Approv, Amendments, & Cert Mgt																					
	b. Technical Standard Order Authorization																					
	c. Parts Manufacturer Approval Design Approvals																					
	d. Conformity Activities																					
	e. Designee Activity																					
	f. Other Design Approval																					
4: Production Approval and Certificate Management	a. Production Approval Holder Supervision																					
	b. Initial Production Approval																					
5: Airworthiness Certification	a. Airworthiness Certification																					
	b. Designee Activity																					
6: Designee Management	a. Supervision/Past Performance																					
	b. Appointment																					

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This matrix shows which components are global and the corresponding support of global sub-functions, by global subcomponent (light shading). Not all sub-functions are supported by all subcomponents due to irrelevancy (dark shading). In addition, the support of the fully-supported sub-functions are detailed by subcomponent. Horizontal hashing indicates ASKME support.

Appendix B: Relevant NTSB cause codes and description**Table B-1: Relevant NTSB cause codes and description**

NTSB Cause Code	Description
82000	Aircraft Equipment Inadequate
82128	Aircraft/Equipment, Inadequate Aircraft Component
82124	Aircraft/Equipment, Inadequate Aircraft Manuals
82126	Aircraft/Equipment, Inadequate Airframe
82100	Aircraft/Equipment, Inadequate Design
82127	Aircraft/Equipment, Inadequate Supplemental Type Certification
24538	Design Stress Limits of Aircraft
70220	Facility, Inadequate Compliance (Documentation)
70210	Facility, Inadequate Compliance (Review)
70220	Facility, Inadequate Compliance Determination
70110	Facility, Inadequate Design
70124	Facility, Inadequate Manuals/Directives
92000	Inadequate Certification/Approval
92200	Inadequate Certification/Approval, Aircraft
92400	Inadequate Certification/Approval, Manufacturer
84110	Inadequate Quality Control
93000	Inadequate Substantiation Process
93300	Inadequate Substantiation Process, Inadequate Documentation
93100	Inadequate Substantiation Process, Inadequate Method of Compliance
93200	Inadequate Substantiation Process, Insufficient Review
90000	Inadequate Surveillance of Operation
90200	Inadequate Surveillance of Operation, Inadequate Procedure
90100	Inadequate Surveillance of Operation, Insufficient Staff
35000	Information
35100	Information Insufficient
35200	Information Unavailable
35300	Information Unclear
35320	Information Unclear (Language)
35310	Information Unclear (Phraseology)
91000	Insufficient Standards/Requirements
91200	Insufficient Standards/Requirements, Aircraft
91400	Insufficient Standards/Requirements, Manufacturer
5331	Kit Manufacturer
4123	Manufacturer
84100	Material Defect
84120	Material Defect (Inadequate Quality Control)
84000	Material Inadequate
84200	Material, Inadequate, Improper
80000	Procedure Inadequate
4129	Production/Design Personnel
4146	Supplier/Distributor of Parts (Other than Manufacturer or Maintenance Personnel)

Appendix C: ASKME Ratings of Accidents

Table C-1: ASKME Ratings of Accidents

FAR Part	Date	NTSB Report #	Accident Location	Aircraft Type	Carrier	Fatalities	Aircraft Damage	ASKME Rating	Notes
121	01/31/2000	DCA00MA023	Anapaca Island, CA	DC-9	Alaska Airlines	88	Destroyed	Low (.30)	Aircraft had inadequate certification approval for lubrication intervals.
121	07/17/96	DCA96MA070	East Moriches, NY	B-747	Trans World Airlines	230	Destroyed	Very Low (.15)	CPS Report* data point – Previous fuel tank problem on record; ASKME to assist preventing this precursor through sub-functions 2a, 3a, 5a, 10a, and 10b.
121	09/08/94	DCA94MA076	Aliquippa, PA	B-737	US Airways	132	Destroyed	Low (.30)	CPS Report* data point – Previous uncommanded rudder actions and design problem; ASKME to assist preventing this recurring precursor through sub-functions 2a, 3a, 5a, 10a, and 10b.
121	07/02/94	DCA94MA065	Charlotte, NC	DC-9	US Airways	37	Destroyed	Low (.30)	Wind shear detection system was sensitive enough to detect the encountered wind shear. However, a feature of the system was to reduce nuisance alerts by desensitizing the detection system during certain periods of the flight. One such time was during flap transition, which is when the accident occurred. The system was not designed to detect wind shear during while the flaps are in transition. Evaluation of STC design data also revealed that the system was not evaluated during a period of flight when the flaps are in transition. ASKME to assist preventing this precursor through sub-functions 2a and 10b.
121	03/03/91	DCA91MA023	Colorado Springs, CO	B-737	United Airlines	25	Destroyed	Low (.30)	CPS Report* data point – Previous flight control rudder occurrences on similar aircraft type. ASKME to assist preventing this precursor through sub-functions 2a, 3a, 5a, 10a, and 10b.
121	02/17/91	DCA91MA021	Cleveland, OH	DC-9	Ryan International	2	Destroyed	Very Low (.15)	ASKME to assist preventing this “Icing” precursor through sub-functions 2a, 3a, 10a, and 10b.
121	07/19/89	DCA89MA063	Sioux City, IA	DC-9	United Airlines	111	Destroyed	Very Low (.15)	CPS Report* data point - Manufacturing flaw - Trend Analysis for repeating similar problems required to prevent this precursor. ASKME to assist through sub-functions 2a, 3a, 4a, 5a, 10a, and 10b.
121	03/18/89	DCA89MA035	Saginaw, TX	DC-9	Evergreen International	2	Destroyed	Low (.30)	ASKME to alert engineers in type certification to upgrade the certification basis to say a cargo door that does not latch/lock properly should not be certified. ASKME to assist preventing this precursor through sub-functions 1a, 2a, 3a, 10b.
121	02/24/89	DCA89MA027	Honolulu, HI	B-747	United Airlines	9	Substantial	Low (.30)	Communication of AD's; AD was not interpreted to continue inspecting cargo doors---only the initial inspection was performed. ASKME to assist preventing this precursor through sub-functions 1a, 2a, 3a, 10b.
129	04/06/93	DCA93MA037	Shanghai, China	MD-11	China Eastern Airlines	2	Minor	Medium (.45)	Information will be captured as part of Lessons learned w/in ASKME to impact future certification.
135	04/27/99	FTW99FA123	Goldsby, OK	CE-402	Texas Air Charters	1	Destroyed	Very Low (.15)	Reason for accident: a recurring defect (wing fell off), which ASKME can proactively correct.
135	07/05/97	ANC97FA099	Skwentia, AK	DH-2	Alaska Bush Carriers	4	Substantial	Very Low (.15)	Partial loss of engine power due to push rod fatigue; A recurring problem with aircraft avoidable with ASKME.
135	07/03/97	ANC97FA097	Skagway, AK	PA-32	Haines Airways	4	Destroyed	Low (.30)	ASKME to provide increased AD's compliance through lessons learned

Fully Releasable

Table C-1: ASKME Ratings of Accidents, Cont'd

135	09/11/92	ANC92FA164	Eagle, AK	HU-369	Tundra Copters	3	Destroyed	Low (.30)	ASKME will deliver the AD and enforce compliance; improve communication with AFS; Part 39 rule change -- ASKME would facilitate the AD; Maintenance unaware of AD existence.
91	07/18/98	FTW98FA325	West Jordan, UT	GA-112-114	Cazier Excavation	1	Destroyed	Very Low (.15)	
91	04/04/96	NYC96GA083	Yeager, WV	BHT-206	West Virginia State Police	2	Destroyed	High (.60)	220 previous reports of the failed area went unaddressed by the Air Cert and Manuf.
91	04/24/90	CHI90FA117	West Bend, WI	Champ-8	Gran Aire, Inc.	2	Destroyed	Very Low (.15)	ASKME would provide information to the PMA part supplier
91	05/09/85	MKC85FA100	Calwood, MO	BHT-47	Dick Hill Helicopters	2	Destroyed	Low (.30)	Possible repeatable issue resolved through ASKME.
91	05/18/83	FTW83FA238	Venice, LA	AS-350	Petroleum Helicopters	1	Destroyed	Low (.30)	
91	09/25/83	NYC83FA256	Syracuse, NY	Mooney-20-F	William D. Stewart	1	Destroyed	Low (.30)	Deteriorated/Inadequate Fuel System; Identify inadequate/poorly designed fuel cap/tank.

Appendix D: Productivity Benefits by Confidence Level and Year**Table D-1: Productivity Benefits by Confidence Interval and Year (In FTEs)**

FY	0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
2006	0.8	1.0	1.0	1.0	1.1	1.1	1.1	1.2	1.2	1.3	1.5
2007	3.3	4.0	4.2	4.4	4.5	4.6	4.8	4.9	5.0	5.3	6.1
2008	8.1	9.9	10.3	10.7	11.0	11.3	11.6	11.9	12.2	12.7	14.6
2009	29.7	35.8	37.3	38.3	39.3	40.2	41.1	42.1	43.2	44.8	51.1
2010	49.0	58.9	61.5	63.5	65.1	66.7	68.4	70.1	72.1	74.8	84.9
2011	67.1	80.4	84.2	87.1	89.4	91.6	94.0	96.6	99.5	103.4	116.7
2012	68.5	82.1	85.9	88.9	91.3	93.5	96.0	98.6	101.6	105.6	119.2
2013	70.0	83.8	87.7	90.7	93.2	95.5	98.0	100.7	103.7	107.8	121.7
2014	71.4	85.5	89.6	92.6	95.1	97.5	100.0	102.8	105.9	110.0	124.2
2015	72.9	87.3	91.4	94.6	97.1	99.5	102.1	104.9	108.1	112.3	126.8
2016	74.5	89.1	93.3	96.5	99.1	101.6	104.3	107.1	110.4	114.7	129.5
2017	76.0	91.0	95.3	98.6	101.2	103.7	106.4	109.4	112.7	117.1	132.2
2018	77.6	92.9	97.3	100.6	103.3	105.9	108.7	111.6	115.0	119.5	134.9
2019	79.2	94.8	99.3	102.7	105.5	108.1	110.9	114.0	117.5	122.0	137.7
2020	80.9	96.8	101.4	104.9	107.7	110.4	113.3	116.4	119.9	124.6	140.6
2021	82.6	98.9	103.5	107.1	109.9	112.7	115.6	118.8	122.4	127.2	143.6
Total	911.6	1092.2	1143.2	1182.2	1213.8	1243.9	1276.3	1311.1	1350.4	1403.1	1585.3